# Letter Sequences & Alphabet Code

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

**Worked examples — tutor talks through these first**

**A) Easy:** What comes next in the sequence: B, D, F, H, ?

*Method & answer: Each letter jumps forward 2 places (B, D, F, H = +2 each time). H + 2 = J. Answer: J.*

**B) Standard:** If the code for CAT is DBU (each letter shifts forward by 1 in the alphabet), what is the code for DOG?

*Method & answer: Shift every letter of DOG forward by 1: D→E, O→P, G→H. Answer: EPH.*

**C) Advanced:** What comes next in the sequence: A, C, F, J, ?

*Method & answer: The gap grows by one each time: +2, +3, +4, so the next gap is +5. A(1)→C(3)→F(6)→J(10)→next = 10+5 = 15 = O. Answer: O.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What comes next in the sequence: A, B, C, D, ?
2. What comes next in the sequence: J, K, L, M, ?
3. What comes next in the sequence: E, G, I, K, ?
4. What comes next in the sequence: C, F, I, L, ?
5. What comes next in the sequence: Z, Y, X, W, ?
6. What comes next in the sequence: T, R, P, N, ?
7. Which letter is 3 letters after F?
8. Which letter is 4 letters before K?
9. What comes next in the sequence: A, C, E, G, ?
10. What comes next in the sequence: M, N, O, P, ?
11. If the code for CAT is DBU (each letter shifts forward by 1), what is the code for DOG?
12. What comes next in the sequence: E, H, K, N, ?
13. What comes next in the sequence: Y, W, U, S, ?
14. Which letter is exactly halfway between B and F in the alphabet?
15. What comes next in the sequence: H, J, L, N, ?

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What comes next in the sequence: X, A, D, G, ?
2. What comes next in the sequence: D, A, X, U, ?
3. If the code for MOUSE is RTZXJ (each letter shifts forward by 5, wrapping round the alphabet), what is the code for HORSE?
6. Which letter is 4 letters after the 24th letter of the alphabet (wrapping round if needed)?
7. What comes next in the sequence: V, Y, B, E, ?
8. In a code, every letter shifts forward by 4, wrapping round the alphabet. The coded word is KSPH. What is the original word?
11. Using a reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X…), what does MOUSE become?
12. Using the same reverse-alphabet code, what does NLFHV decode back to?
13. What comes next in the sequence: W, B, G, L, ?

**4.** Which letter is 9 letters after R, wrapping round the alphabet if needed?

**5.** Which letter is 10 letters before C, wrapping round the alphabet if needed?

**9.** What comes next in the sequence: A, C, F, H, K, ? (the gaps alternate +2, +3)

**10.** Which letter is exactly halfway between E and Q in the alphabet?

**14.** If A=1, B=2, C=3 ... Z=26, and the count continues past Z by wrapping back round to A (so 27=A, 28=B...), which letter is represented by the number 30?

**15.** Using the reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X...), which letter becomes J after the swap?

## Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** What comes next in the sequence: A, Z, B, Y, C, ?

**2.** What comes next in the sequence: B, D, G, K, ?

**3.** What comes next in the sequence: X, Z, B, D, ?

**4.** If the code for PLUM is RNWO (each letter shifts forward by 2), what is the code for PEAR?

**5.** Using a code where every letter shifts back by 2 (each letter becomes the letter 2 earlier), the coded word is KGAC. What is the original word?

**6.** What comes next in the sequence: A, C, E, H, K, ? (the gap changes partway through)

**7.** A code reverses the letters of a word, then shifts each one forward by 1. Applying this to CAT gives UBD (reverse to TAC, then T→U, A→B, C→D). Using the same rule, what does DOG become?

**8.** In a code, the 1st letter of a word shifts forward by 1, the 2nd letter by 2, and the 3rd letter by 3. Using this rule, SUN becomes TWQ (S+1=T, U+2=W, N+3=Q). What does the same rule turn CAB into?

**9.** What comes next in the sequence: A, Y, C, W, E, ?

**10.** Which letter is the same number of places from Z as D is from A?

**11.** In a code, two letters are added by adding their alphabet positions: the code for 'B+D' is F (2+4=6=F). Using the same rule, what single letter represents 'E+G'?

**12.** A code shifts consonants forward by 1 and vowels backward by 1 (wrapping from A back to Z). Using this rule, CAT becomes DZU (C→D, A→Z, T→U). What does the same rule turn DOG into?

**13.** A code shifts every letter forward by 3, wrapping from Z back to A. Using this code, YB becomes BE (Y→B, B→E). What does ZC become under the same code?

**14.** What comes next in the sequence: A, B, C, E, H, ? (the gaps follow 1, 1, 2, 3...)

**15.** A code first shifts a letter forward by 1, then swaps it using the reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X...). Applying this to A gives: A+1=B, then reverse of B is Y, so A→Y. Using the same two-step rule, what does C become?